AI531C — Week 13 Pre-Class Activity (PCM-13: Score the Mask)

Prepares: Lectures 37–38 (online window 5–11 Nov: Segmentation evaluation; Change detection) · Lecture 39 (Siamese, Tue 10 Nov) · Lectures 40–41 (Fusion; Responsible use), Wed 11 Nov · Tutorial T13 (Tue 10 Nov, metrics clinic).

1 · Topic

Dice, IoU and the 99%-accuracy trap — the metrics week, delivered partly online. This card is the T13 clinic's entry ticket.

2 · Why you are preparing this

L37–L38 run as recordings in the make-up window; the T13 clinic then scores YOUR masks live. Doing one Dice/IoU by hand first means the clinic debugs your numbers instead of teaching the formulas.

3 · Read (12–15 min)

From the Comprehensive Self-Contained Lecture Notes:

– Lecture 37 chapter, the sections on Dice/IoU and on class imbalance (about 3 pages);

– Lecture 38 chapter, ONLY the two-assumptions page (registration + photometric stability).

Optional depth: watch the L37 recording's first ten minutes before attempting item 5.

4 · Interpret one figure (4 min)



Source: course figure, Lecture 37 (toy overlap; the imbalance trap).

In two sentences: the right panel shows a model scoring 99% accuracy — what did it actually find, and which bar exposes the fraud?

5 · One small calculation (5 min)

Ground truth |A| = 80 px, prediction |B| = 60 px, overlap 48 px. Compute Dice and IoU (three decimals), then verify IoU = Dice/(2−Dice).

6 · Predict (4 min)

Predict: your all-background 'model' on a 2%-foreground image — accuracy, Dice, sensitivity (three numbers, no computing needed). Which one lies?

7 · Inspect this fragment (3 min)

```
tp = (pred & gt).sum(); fp = (pred & ~gt).sum()
fn = (~pred & gt).sum()
dice = 2*tp / (2*tp + fp + fn)
```

Question: true negatives appear nowhere in this code. Why is that the POINT rather than an omission?

8 · Bring one question to class

Your own masks (or your change maps) with their three counts (TP/FP/FN) precomputed — T13 scores them live. Plus one question from the L37/L38 recordings for the live Q&A.

Submit: items 4–6 — half a page maximum, before Tue 10 Nov, 9:00 AM. Annotation/change cards due at T13.

Expected time: 30–35 minutes. Counts toward the weekly preparation portfolio (best 10 of PCM-2…PCM-14, completion-based micro-credit per the lesson plan). The short per-lecture primers in the course folder are optional warm-ups; only THIS card is submitted.